CARAT: Class-Aware Robust Aggregation for Model-Poisoning-Resilient Federated Learning

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Abstract

Federated learning is vulnerable to model-poisoning attacks because the server must aggregate client updates without inspecting client data. Existing robust aggregation rules often rely on update geometry, coordinate-wise robust statistics, or a single trusted reference direction, but these signals can confuse malicious behavior with benign client heterogeneity under non-i.i.d. data. We propose **CARAT**, a class-aware robust aggregation rule that uses a small labeled server reference set to sample hidden, class-balanced tasks and score each client update by its task-level loss reduction. CARAT combines these hidden-task certificates with robust clipping, common-radius evaluation, coordinate-rank anomaly penalties, and capped-simplex weight optimization to favor updates that remain useful across semantic slices while limiting structurally extreme clients. On CIFAR-100/ResNet18 with 20% malicious clients, two non-i.i.d. regimes, and four untargeted model-poisoning attacks, CARAT achieves the best average rank among the reported public baselines and improves final accuracy by 1.9–5.5 percentage points over the strongest non-CARAT baseline in each attack/heterogeneity setting. Within this scoped evaluation, the results show that task-conditioned validation is a practical alternative to purely geometric or single-anchor trust signals when a modest labeled server audit set is available.

1 Introduction

Federated learning (FL) enables many clients to train a shared model without centralizing their private data [1–3]. This distributed training paradigm is attractive for privacy- and governance-sensitive applications, but it also makes aggregation a security-critical operation. At each round, the server must update the global model from client messages whose underlying data and training process cannot be directly inspected. As a result, the aggregation rule becomes the main line of defense when some clients submit harmful updates.

Model-poisoning attacks exploit this trust bottleneck by replacing benign client updates with crafted messages. These attacks can be targeted, such as backdoor and model-replacement attacks [4–6], or untargeted, where the goal is to degrade clean test performance and destabilize collaborative training [7–9]. This paper focuses on the untargeted setting. The core challenge is not only that malicious updates can be arbitrary, but also that benign updates are naturally diverse under non-i.i.d. client data. A geometrically unusual update may therefore be either an attack or a legitimate consequence of client heterogeneity [10–13].

Existing Byzantine-robust defenses mitigate this threat through geometric selection, coordinate-wise robust statistics, or auxiliary validation, but their trust signals are often too coarse for heterogeneous FL. Krum and MultiKrum [14] select centrally located updates, Bulyan [15] combines selection with a second robust aggregation stage, coordinate-wise median [16] suppresses marginal outliers, and RFA

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[17] uses a geometric-median objective. These rules can confuse benign dispersion with malicious deviation when client data are non-i.i.d. Auxiliary-information defenses partly address this limitation, but FLTrust [18] compresses trusted data into a single reference direction and FLDetector [19] relies on historical behavior prediction. Such signals may miss attacks whose harm is concentrated on a subset of classes or semantic regions. The resulting open problem is how to validate client updates by their task-level usefulness rather than only by their location in update space.

We propose **CARAT**, a class-aware robust aggregation rule that validates updates through hidden, class-balanced tasks sampled from a small labeled server reference set. The key idea is that a benign non-i.i.d. update need not be geometrically central, but it should provide useful evidence on semantic validation tasks that are hidden from the clients. CARAT first places candidate updates in a common scoring geometry through robust clipping and common-radius evaluation. It then measures task-level loss reduction to form hidden-task certificates, penalizes coordinate-rank anomalies, and solves a capped-simplex weight optimization problem with temporal regularization. The final update is computed from clipped client updates, so the server can preserve useful benign diversity while limiting clients that are semantically harmful or structurally extreme. This design explicitly targets settings where a modest labeled server audit set is available; it is not intended to be a fully data-free defense.

We evaluate CARAT on CIFAR-100/ResNet18 under four untargeted model-poisoning attacks, two Dirichlet non-i.i.d. regimes, and a 20% malicious-client ratio. CARAT achieves the best average rank among the reported public baselines and improves final accuracy by 1.9–5.5 percentage points over the strongest non-CARAT baseline across the tested attack/heterogeneity settings. These results support a deliberately bounded claim: class-conditioned validation can provide a practical robustness signal that outperforms purely geometric or single-anchor criteria within the completed reference-assisted evaluation scope. Our main contributions are as follows.

- We introduce hidden-task certificates for robust FL aggregation, using class-balanced server-side validation tasks to evaluate whether each client update improves semantic slices of the learning problem.
- We develop the CARAT aggregation objective, which combines hidden-task utility, coordinate-rank anomaly penalties, capped per-client influence, and temporal regularization into a single robust weighting rule.
- We provide an auditable CIFAR-100/ResNet18 evaluation against citable robust-aggregation baselines under four untargeted poisoning attacks and non-i.i.d. partitions, with matched server-reference budgets for reference-assisted defenses and seed-level reporting.

2 Related Work

Model-poisoning attacks. Model-poisoning attacks directly manipulate client model updates before aggregation; in this paper we focus on untargeted availability attacks that prevent convergence or degrade clean test performance. Early Byzantine and distributed-learning work studied large-noise attacks in Byzantine-tolerant gradient descent [14], sign-flipping attacks in asynchronous Byzantine SGD [20], and local model manipulation in federated learning [21], establishing that robust aggregation must tolerate arbitrary client messages rather than only stochastic training noise. Later attacks became substantially more adaptive. ALIE [7] exploits benign gradient variance to remain statistically plausible, IPM [22] reverses progress through inner-product manipulation, FangAttack [8] optimizes local model poisoning against specific robust aggregators, and MinMax/MinSum [9] search for malicious updates that satisfy distance- or sum-based stealth constraints while damaging the global model. Mimic-style attacks [23] further show that heterogeneity itself can be weaponized by making poisoned updates resemble legitimate client drift. Targeted backdoor and model-replacement attacks form an important adjacent threat family, including model replacement [4], distributed backdoor attacks [5], and tail-class backdoors [6], but CARAT's empirical claim is deliberately limited to untargeted model poisoning.

Defenses against model poisoning. Defenses against model poisoning differ mainly in the trust signal used for aggregation. Classical Byzantine-robust rules rely on geometry or coordinate-wise statistics: Krum and MultiKrum [14] select centrally located updates, Bulyan [15] combines selection with a second robust aggregation stage, coordinate-wise median [16] suppresses marginal outliers,

and RFA [17] uses a geometric-median objective. Subsequent defenses add suspicion scoring through Zeno [24], redundant gradients through DRACO [25], regularized aggregation through RSA [26], divide-and-conquer filtering through DnC [9], history-aware clipping through CenteredClipping [27], sign-pattern filtering through SignGuard [28], latent-space screening through FLARE [29], and layer-adaptive sparsification through LASA [30]. Because non-i.i.d. data can make benign updates geometrically dispersed, another line explicitly studies robustness under heterogeneity through FedProx [11], SCAFFOLD [12], resampling [31], bucketing [23], local-SGD analysis [32], gradient splitting [33], nearest-neighbor mixing [34], label-skew-aware aggregation [35], and entropy-guided public-data aggregation [36]; non-i.i.d. FL surveys provide the broader context [13]. Auxiliary-information defenses use trusted data, historical behavior, or diagnostic signals rather than only instantaneous update geometry: AUROR [37] screens updates in collaborative learning, FLTrust [18] bootstraps trust from a server reference update, FoolsGold [38] penalizes repeated sybil-like similarity, and FLDetector [19] predicts suspicious client behavior from history. CARAT follows this auxiliary-information direction but changes the validation principle: instead of trusting one anchor direction, one geometry statistic, or one historical detector, it asks whether each update remains useful across multiple hidden, class-balanced semantic tasks under a matched server-reference budget.

3 Preliminaries and Problem Statement

3.1 Federated Learning and Threat Model

We study synchronous FL in which the server must update a global model from client messages without inspecting client data. Let $\mathcal{N} = \{1, \ldots, N\}$ be the client population. Client i holds local data $\mathcal{D}_i$ and objective $F_i(w)$, giving the standard weighted objective

$$F(w) = \sum_{i=1}^{N} p_i F_i(w), \qquad p_i = \frac{|\mathcal{D}_i|}{\sum_{j=1}^{N} |\mathcal{D}_j|}. \tag{1}$$

At communication round r, the server broadcasts w_r to selected clients $\mathcal{S}_r$. Each selected client initializes $w_{i,r}^0 = w_r$ and performs K local stochastic-gradient steps,

$$w_{i,r}^{k+1} = w_{i,r}^k - \eta_r \nabla F_i(w_{i,r}^k; \xi_{i,r}^k), \qquad k = 0, \ldots, K-1. \tag{2}$$

The returned model message is converted to an update vector

$$\Delta_i^r = w_{i,r}^K - w_r, \tag{3}$$

and a server-side aggregation rule maps the received updates to the next global model,

$$w_{r+1} = w_r + \mathcal{A}\big(\{\Delta_i^r : i \in \mathcal{S}_r\}\big). \tag{4}$$

FedAvg is recovered when $\mathcal{A}$ is the sample-size-weighted mean.

The adversary controls a subset $\mathcal{M}_r \subseteq \mathcal{S}_r$ of selected clients and may replace their benign updates with arbitrary crafted messages $\widetilde{\Delta}_i^r$. We focus on untargeted model poisoning: malicious updates aim to degrade clean test accuracy while remaining plausible enough to evade robust aggregation. This setting is difficult under non-i.i.d. data because benign clients can naturally differ in update direction, norm, sparsity, and class emphasis. A robust aggregator must therefore reduce harmful influence without treating every geometrically atypical update as malicious client behavior.

The adversary is assumed to know the training protocol, defense family, and public hyperparameters, but not the realized hidden validation tasks sampled by the server in a given round. This is the standard security boundary for CARAT's reference-assisted setting: the labeled reference set and task sampler are server-side audit resources, not public client inputs. Attackers that directly observe the reference examples or optimize against the exact hidden tasks are outside the empirical claim set, although the method still includes clipping, rank penalties, and capped weights to reduce overreliance on the reference signal.

3.2 Reference-Assisted Robust Aggregation

CARAT assumes that the server maintains a small labeled reference set $\mathcal{D}^{\mathrm{ref}}$ covering the task label space. From this data it constructs a probe pool $\mathcal{P}$ and samples hidden, class-balanced validation

tasks during aggregation. This is a stronger setting than fully data-free robust aggregation and should be treated as part of the threat model. The distinction from single-anchor trusted-data methods is that CARAT does not compress $\mathcal{D}^{\text{ref}}$ into one trusted update direction; instead, it uses the reference data to instantiate multiple semantic checks that are hidden from clients.

Given the round-level update set $\{\Delta_i^r\}_{i \in \mathcal{S}_r}$ and reference data $\mathcal{D}^{\text{ref}}$, the server must construct nonnegative aggregation weights and an aggregate update that preserve useful benign diversity while suppressing poisoned clients. This goal imposes four design requirements. First, update scoring should be scale-controlled so that large-norm messages cannot win merely by magnitude. Second, trust should depend on task-conditioned utility, not only on distance to other clients. Third, the method should still penalize structurally extreme updates, because a harmful client may look useful on part of the probe signal. Finally, the aggregation weights should have bounded per-client influence and temporal stability across rounds. CARAT implements these requirements through common-geometry scoring, hidden-task certificates, coordinate-rank penalties, and capped-simplex optimization.

4 CARAT

4.1 Overview

CARAT turns the requirements above into the three-stage server-side pipeline shown in Figure 1. First, the server constructs hidden class-balanced tasks from a labeled reference set, making the validation queries semantic and unavailable to clients. Second, CARAT evaluates each client update on these hidden tasks after placing updates in a common scoring geometry, producing a certificate matrix of task-level loss reductions. Third, CARAT learns bounded aggregation weights by optimizing worst-task utility under rank-anomaly, temporal-stability, and per-client influence constraints. The final server update is a weighted average of clipped client updates, not of the normalized evaluation copies.

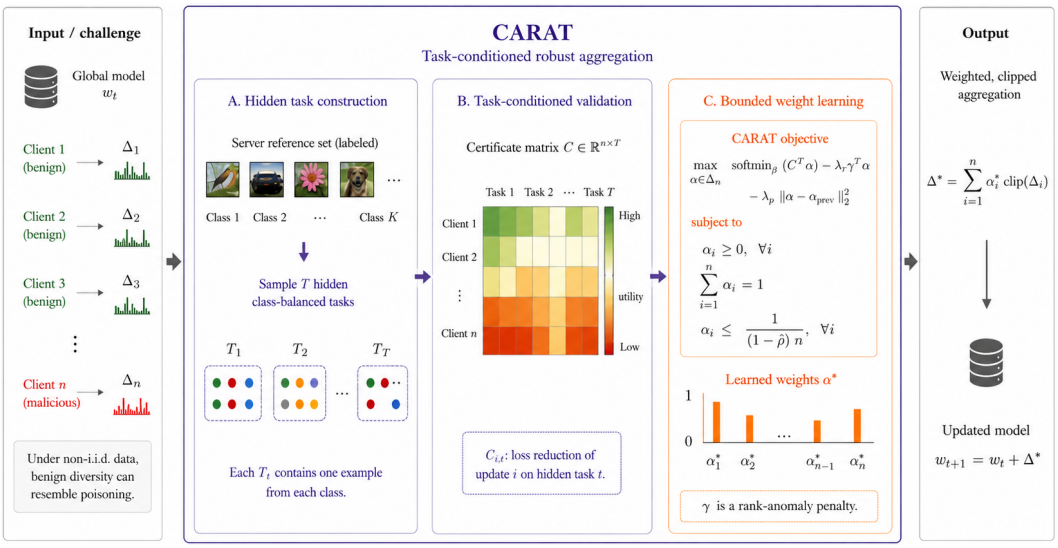

Figure 1: Overview of CARAT. Under non-i.i.d. data, benign diversity can resemble poisoning. CARAT addresses this ambiguity through three server-side stages: (A) hidden class-balanced task construction from a labeled reference set, (B) task-conditioned validation that forms a certificate matrix of loss reductions, and (C) bounded weight learning with rank-anomaly, temporal-stability, and per-client cap constraints. The server updates the model using a weighted average of clipped client updates.

Appendix A.3 summarizes how each component maps to a concrete failure mode in non-i.i.d. robust aggregation.

4.2 Hidden Tasks and Task-Conditioned Validation

Within a round, CARAT first prepares client updates for task-conditioned validation. We suppress the round superscript in this section. Before semantic evaluation, the server clips each update so that a small number of large-magnitude messages cannot dominate the comparison stage. Let $\|\Delta_i\|_2$ denote the norm of client i's update. In our implementation, the clipping threshold τ_r is estimated by the MAD-based rule

$$\tau_r = \mathrm{median}(\{\|\Delta_i\|_2\}) + k \cdot 1.4826 \cdot \mathrm{MAD}(\{\|\Delta_i\|_2\}), \tag{5}$$

and the clipped update is

$$\bar{\Delta}_i = \min\left(1, \frac{\tau_r}{\|\Delta_i\|_2 + \varepsilon}\right) \Delta_i. \tag{6}$$

CARAT then rescales each clipped update to a common evaluation radius q_r,

$$\tilde{\Delta}_i = q_r \frac{\bar{\Delta}_i}{\|\bar{\Delta}_i\|_2 + \varepsilon}, \tag{7}$$

where q_r is chosen as a quantile of the positive clipped norms. The normalized vector $\tilde{\Delta}_i$ is used only to evaluate candidate utility; the final aggregation step uses $\bar{\Delta}_i$. This separation makes certificate scoring scale-neutral without changing the conservative clipped update applied to the global model.

CARAT then constructs hidden semantic validation tasks from the server reference data. It samples T hidden tasks from the probe pool $\mathcal{P}$, with each task constructed to be approximately class-balanced. This class-balanced construction is the source of CARAT's class awareness: a malicious update may remain globally plausible while degrading only a subset of classes, whereas repeated balanced tasks make localized failures harder to hide. For task t, let $\ell_t(w)$ denote the average cross-entropy loss of model w on that task. CARAT defines a *hidden-task certificate*

$$c_{i,t} = \ell_t(w_r) - \ell_t(w_r + \tilde{\Delta}_i). \tag{8}$$

Here, "certificate" denotes an internal task-level utility score, not a formal certified-robustness guarantee. Positive values indicate that client i improves task t relative to the current global model.

Stacking these scores yields a certificate matrix $C \in \mathbb{R}^{n \times T}$. Because tasks may have different intrinsic difficulty and baseline loss scale, CARAT robustly standardizes each column of C before optimization. This preserves within-task ordering while making scores comparable across hidden slices. The matrix gives the server a richer signal than client-to-client similarity alone: a client should be trusted because it remains broadly useful across hidden tasks, not merely because it lies near the geometric center of the client population.

Task utility alone is insufficient, because a malicious client may overfit the probe tasks or align with part of the semantic signal while remaining structurally extreme in parameter space. CARAT therefore augments hidden-task certificates with a coordinate-rank anomaly score. For a sampled coordinate set $S \subseteq \{1, \ldots, d\}$, it ranks clients coordinate-wise and measures the average normalized distance from the center rank:

$$\gamma_i^{(S)} = \frac{1}{|S|} \sum_{j \in S} \left| \frac{\mathrm{rank}_{i,j}}{n-1} - \frac{1}{2} \right|. \tag{9}$$

CARAT repeats this computation over several random coordinate subsets and aggregates the resulting values robustly. After robust standardization, only positive deviations are retained as the rank-penalty vector $\gamma \in \mathbb{R}^n$. This signal is intentionally asymmetric: suspicious extremeness is penalized, whereas unusual centrality is not explicitly rewarded. Together, the certificate matrix and rank penalty separate two notions that are often conflated in robust aggregation: whether an update is useful and whether it is structurally atypical.

4.3 Bounded Weight Learning

CARAT maps task-conditioned validation signals to aggregation weights through a bounded optimization problem on the capped simplex

$$\mathcal{W}(\hat{\rho}) = \left\{ \alpha \in \mathbb{R}^n : \alpha_i \geq 0,\ \sum_{i=1}^{n} \alpha_i = 1,\ \alpha_i \leq \frac{1}{(1-\hat{\rho})n} \right\}, \tag{10}$$

where $\hat{\rho}$ is an estimate of the attacker fraction. The cap prevents any single client from dominating the aggregate and encodes a conservative trust budget at the weight level: even when a client appears useful, it cannot absorb arbitrary influence within one round.

Let α^{prev} be the previous round's weights. CARAT uses the quadratic temporal-variation penalty $\|\alpha - \alpha^{\text{prev}}\|_2^2$ and solves

$$\max_{\alpha \in \mathcal{W}(\hat{\rho})} \operatorname*{softmin}_{\beta}(C^\top \alpha) - \lambda_r \gamma^\top \alpha - \lambda_p \|\alpha - \alpha^{\text{prev}}\|_2^2, \tag{11}$$

where

$$\operatorname*{softmin}_{\beta}(s_1, \ldots, s_T) = -\frac{1}{\beta} \log \left(\frac{1}{T} \sum_{t=1}^{T} e^{-\beta s_t} \right). \tag{12}$$

The first term is a smooth approximation to worst-task utility: large β places more emphasis on the weakest hidden tasks, while finite β preserves optimization stability. The second term penalizes structurally extreme clients, and the third term regularizes the solution toward the previous round to reduce unnecessary oscillation. Eq. (11) implements a simple decision rule: a client should retain influence only if it remains useful across hidden tasks and does not repeatedly appear as an outlier in rank space.

After optimization, CARAT aggregates the clipped updates:

$$\Delta_{\text{agg}} = \sum_{i=1}^{n} \alpha_i \bar{\Delta}_i. \tag{13}$$

This is the final server step. The aggregate inherits the robustness benefits of clipping while using weights learned from the richer evaluation geometry defined above. In practice, the main additional computation comes from evaluating n candidate updates over T hidden tasks and computing rank penalties on coordinate subsamples. Probe-model forward passes dominate, while the ranking and optimization stages remain tunable through explicit budget parameters.

A compact round-level algorithm summary is provided in Appendix A.2. Structural properties of feasible weighting, clipped aggregation, malicious-weight influence, hidden-task evaluation geometry, objective behavior, and per-round complexity are deferred to Appendix A.4.

5 Experiments

The experiments evaluate one deliberately scoped claim: whether hidden-task certificates improve robustness to untargeted model poisoning under heterogeneous client data, when compared against public robust-aggregation baselines under a fixed paper protocol and an explicit result-source policy. The main study uses CIFAR-100/ResNet18, matched trusted-data budgets for CARAT and FLTrust, final attacked accuracy/loss metrics, and average rank across attacks. Appendix B provides protocol details, final-loss results, runtime diagnostics, and reproducibility details.

5.1 Evaluation Protocol

Benchmark and threat model. We use CIFAR-100 with ResNet18 [39]. CIFAR-100 has a class-rich label space, making it suitable for evaluating whether hidden class-balanced tasks reveal class-selective degradation that may be hidden in aggregate accuracy alone. The attacked comparison covers four untargeted model-poisoning attacks: ALIE [7], FangAttack [8], MinMax [9], and MinSum [9]. These attacks test stealthy variance-based poisoning, geometry-aware local model poisoning, and distance-constrained optimization attacks. Backdoor and data-poisoning attacks remain outside the empirical claim set.

Baselines and fairness controls. The main-text comparison includes Mean, NormClipping, Multi-Krum [14], FLTrust [18], FLDetector [19], and CARAT. This set spans averaging, clipping, geometry-based filtering, trusted-data anchoring, historical detection, and hidden-task validation. We exclude unpublished or under-review defenses from the main claim so that every reported baseline is citable and publicly attributable. To isolate the effect of the aggregation rule, CARAT and FLTrust use matched labeled server-reference budgets.

Training and reporting protocol. All main attacked runs use 20 clients with full participation, 200 global rounds, 4 local epochs, learning rate 0.05, and a 20% malicious-client ratio. We evaluate Dirichlet non-i.i.d. regimes with $\alpha \in \{1, 0.5\}$, where smaller α increases client heterogeneity. Clean no-attack runs are kept separate from attacked robustness tables so that clean utility is not mixed with poisoning resistance. Compatible legacy public-baseline rows use five seeds where available; regenerated FLTrust and CARAT rows use three seeds under the matched reference-assisted protocol. We disclose this source policy in the tables and use average rank to avoid overinterpreting any single seed count. Final test accuracy is the primary metric, final test loss in the appendix is a consistency check, and average rank within each non-i.i.d. setting summarizes performance across attacks.

Table 1: Reported evaluation protocol used for the main CIFAR-100/ResNet18 robustness comparison. The table separates training controls, poisoning controls, and reporting controls.

Axis	Setting	Purpose in the comparison
Benchmark and federated training protocol		
Dataset / model	CIFAR-100 / ResNet18	Class-rich image benchmark where hidden class-balanced tasks are meaningful.
Clients	20 clients, full participation	Removes client-sampling noise from defense comparisons.
Training schedule	200 rounds; 4 local epochs; learning rate 0.05	Shared optimizer budget for all attacked runs.
Heterogeneity	Dirichlet non-i.i.d. $\alpha \in \{1, 0.5\}$	Tests robustness as benign client updates become more dispersed.
Poisoning setting and compared defenses		
Malicious clients	20% of clients	Main untargeted model-poisoning threat level.
Attacks	ALIE, FangAttack, MinMax, MinSum	Stealthy and geometry-aware untargeted poisoning attacks.
Main-text defenses	Mean, NormClipping, MultiKrum, FLTrust, FLDetector, CARAT	Public citable baselines spanning averaging, clipping, geometric filtering, trusted anchoring, and historical detection.
Reporting controls		
Metrics	Final test accuracy; final test loss; average rank	Accuracy is the main robustness metric; loss and rank provide consistency checks.
Trusted-data control	Matched labeled reference budget for FLTrust and CARAT	Prevents the proposed method from receiving more server-side supervision.
Seed policy	Legacy public-baseline rows: five seeds where available; regenerated FLTrust/CARAT rows: three seeds	Makes the result-source policy explicit while keeping the trusted-data budget matched for reference-assisted defenses.

5.2 Main Robustness Results

Table 2 reports the main attacked-accuracy comparison. The table uses the reported public baselines described above, reuses protocol-compatible five-seed legacy logs for Mean, NormClipping, MultiKrum, and FLDetector, and regenerates matched-budget FLTrust and the final $T = 8$ CARAT protocol with three seeds. The no-attack Mean run is not inserted as an attack-row baseline; it is shown only as a reference curve in Figure 2.

Within this reported comparison, CARAT obtains average rank 1.0 in both heterogeneity settings and is the strongest reported defense in every attack column. Relative to the best non-CARAT row in each column, CARAT improves mean final accuracy by 1.9–5.5 percentage points across the eight attack/heterogeneity cells. The largest margins occur under FangAttack and under the more heterogeneous $\alpha = 0.5$ MinMax/MinSum settings, where several non-CARAT defenses lose substantial accuracy. Appendix B reports the corresponding final-loss table.

Figure 2 shows the round-wise behavior under FangAttack, the attack where several non-CARAT defenses degrade most visibly in Table 2.

Table 2: Final test accuracy (%) for the reported public baselines under four model-poisoning attacks on CIFAR-100/ResNet18 with 20% malicious clients and non-i.i.d. client partitions. Higher accuracy is better, and lower average rank is better. Legacy public-baseline rows reuse five-seed logs where available; FLTrust and CARAT are regenerated under the reference-assisted protocol with three seeds and matched reference budgets.

Dataset Model	Non-IID setting	Defense rule	Attacks				Avg. rank
			ALIE	**FangAttack**	**MinMax**	**MinSum**	
CIFAR-100 ResNet18	$\alpha = 1.0$	Mean	49.43 ± 0.39	48.86 ± 0.70	50.49 ± 0.59	50.50 ± 0.49	2.2
		NormClipping	25.27 ± 0.70	22.67 ± 0.77	24.86 ± 0.65	24.74 ± 0.63	6.0
		MultiKrum	39.59 ± 0.74	37.18 ± 0.83	48.69 ± 0.84	47.71 ± 1.12	4.0
		FLTrust (matched)	39.17 ± 1.87	37.68 ± 1.92	41.90 ± 1.18	41.56 ± 1.61	4.5
		FLDetector	47.96 ± 0.58	28.54 ± 4.42	50.58 ± 0.76	50.47 ± 0.27	3.2
		CARAT ($T = 8$)	51.33 ± 1.75	54.31 ± 1.28	54.35 ± 1.46	54.70 ± 1.43	1.0
CIFAR-100 ResNet18	$\alpha = 0.5$	Mean	47.62 ± 0.73	47.53 ± 1.01	48.88 ± 0.32	48.88 ± 0.52	2.2
		NormClipping	23.73 ± 0.53	20.64 ± 1.41	23.28 ± 0.40	22.92 ± 0.50	6.0
		MultiKrum	37.00 ± 0.81	33.05 ± 1.20	46.59 ± 1.09	46.45 ± 0.80	4.0
		FLTrust (matched)	34.52 ± 0.84	34.22 ± 0.94	39.50 ± 0.27	39.49 ± 0.37	4.5
		FLDetector	45.30 ± 0.75	27.77 ± 1.25	48.85 ± 0.53	49.01 ± 0.69	3.2
		CARAT ($T = 8$)	50.15 ± 1.13	51.81 ± 0.79	53.64 ± 1.12	54.10 ± 0.30	1.0

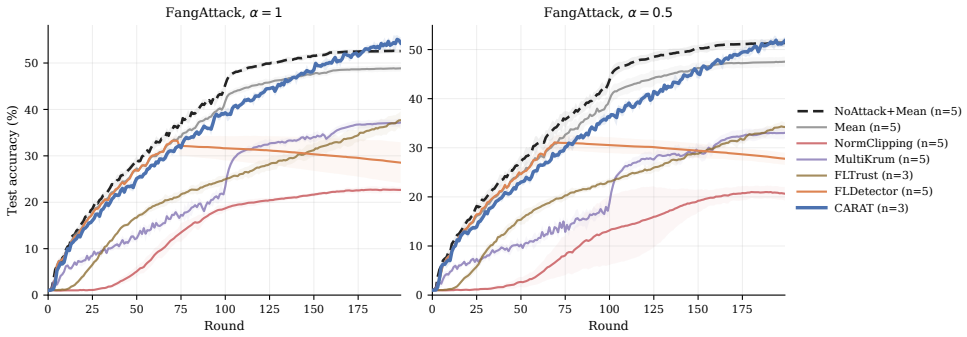

Figure 2: Evolution of test accuracy on CIFAR-100/ResNet18 under FangAttack with 20% malicious clients and non-i.i.d. data. Curves show seed means and shaded bands show one standard deviation. The dashed black curve is the no-attack Mean reference under the same client partition protocol.

5.3 Scope and Diagnostics

The completed evaluation supports a bounded claim: robustness against untargeted model poisoning on CIFAR-100/ResNet18 with 20% malicious clients and Dirichlet non-i.i.d. partitions with $\alpha \in \{1, 0.5\}$. More severe heterogeneity, larger client populations, additional datasets, backdoor attacks, data-poisoning attacks, and attackers that observe the exact hidden validation tasks remain outside the empirical claim set. This boundary is intentional: the paper evaluates whether class-aware hidden-task validation is a useful reference-assisted aggregation signal, not whether CARAT solves all Byzantine FL settings. Since Proposition A.9 identifies probe-model evaluation as CARAT's main extra cost, Appendix B.5 reports verification-scale learning curves and runtime measurements as overhead diagnostics rather than as primary robustness claims.

6 Conclusion

We presented CARAT, a class-aware robust aggregation rule that validates client updates through hidden, class-balanced task certificates rather than relying only on geometric centrality or a single trusted reference direction. Across the completed CIFAR-100/ResNet18 non-i.i.d. evaluation with matched trusted-data budgets for reference-assisted defenses, CARAT achieves the strongest performance against the reported public baselines under four untargeted model-poisoning attacks. These

results support CARAT as a practical defense for FL settings where a modest labeled server audit set is available and class-selective failures matter under heterogeneous client drift.

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Appendix organization. The appendix is organized to make each claim auditable. Appendix A gives the round-level algorithm, implementation invariants, and proofs of the structural properties used in the main text. Appendix B gives only empirical support for the completed claim: protocol details, CARAT hyperparameters, final-loss results, runtime diagnostics, and reproducibility information.

A Algorithmic and Theoretical Details

This appendix provides the technical material needed to reproduce and interpret CARAT. It first fixes the round-level notation and server inputs, then gives executable pseudocode, implementation invariants, and structural propositions for the scoring and weighting rule. These results are deliberately stated as invariants and bounds for the proposed aggregation procedure, not as a complete Byzantine-robustness guarantee against arbitrary adaptive attackers.

A.1 Notation and Round Inputs

All quantities in this appendix refer to one communication round r unless stated otherwise. The server receives $n = |\mathcal{S}_r|$ client messages and converts them into vector updates Δ_i^r. CARAT keeps three versions of each update:

$$\Delta_i^r \quad \text{(raw client update)}, \qquad \bar{\Delta}_i^r \quad \text{(clipped update)}, \qquad \tilde{\Delta}_i^r \quad \text{(common-radius scoring copy)}.$$

Only $\bar{\Delta}_i^r$ is used in the final server update. The normalized copy $\tilde{\Delta}_i^r$ is used only to evaluate hidden-task utility, so the validation score is less sensitive to residual norm differences after clipping.

The server maintains a labeled reference set $\mathcal{D}^{\mathrm{ref}}$ and derives from it a working probe pool $\mathcal{P}$. Each hidden task is a small approximately class-balanced subset sampled from $\mathcal{P}$. The raw certificate matrix is $C^{\mathrm{raw}} \in \mathbb{R}^{n\times T}$, where T is the number of hidden tasks and $C^{\mathrm{raw}}_{i,t} = c_{i,t}$ is the loss reduction obtained by candidate update i on hidden task t. After robust column-wise standardization, the optimization uses $C \in \mathbb{R}^{n\times T}$. The vector $\gamma \in \mathbb{R}^n_{\geq 0}$ denotes coordinate-rank anomaly penalties. The vector α^{prev} is the previous round's aggregation weight vector and acts as a temporal prior.

A.2 CARAT Round Algorithm

Algorithm 1 summarizes one CARAT server aggregation round. The algorithm is server-side: clients do not observe the realized hidden tasks sampled from $\mathcal{P}$.

A.3 Implementation Invariants

This subsection records implementation choices that affect reproducibility but are too low-level for the main method section. The purpose is to make the server-side scoring rule auditable, not to introduce additional assumptions beyond those stated in the threat model.

Component rationale. Table 3 summarizes why each CARAT component is included. The design is not intended as an arbitrary stack of heuristics: each term targets a distinct failure mode that appears when heterogeneous benign drift and malicious manipulation coexist.

Persistent state and initialization. CARAT maintains two state variables across rounds: the probe pool $\mathcal{P}$ derived from $\mathcal{D}^{\mathrm{ref}}$, and the previous-round weight vector α^{prev}. At the first round, α^{prev} is initialized to the uniform vector. This makes the first aggregation unbiased across clients while allowing later rounds to use temporal consistency once weight trajectories have been observed.

Reference-pool construction and hidden-task sampling. Let $\mathcal{P}_c \subseteq \mathcal{P}$ denote the subset of probe examples with class label c. A hidden task is constructed by sampling a small number of examples from each available class pool and concatenating them into one balanced micro-task. If the desired number of samples per class is not available without replacement, CARAT falls back to replacement and, if needed, to a random subset of the probe pool rather than aborting the round. This fail-soft design preserves the intended semantic diversity of the task sampler while ensuring that the server always obtains a well-defined certificate matrix.

Algorithm 1 CARAT server aggregation at round r

Require: Current global model w_r, client messages $\{u_i^r\}_{i=1}^n$, previous-round weights α^{prev}, server reference set $\mathcal{D}^{\text{ref}}$, probe pool $\mathcal{P}$, CARAT hyperparameters
Ensure: Aggregated update Δ_{agg}, updated temporal prior α^{prev}

```
if P is uninitialized then
    Construct a class-balanced probe pool P from D^ref
end if
if r = 1 then
    α^prev ← (1/n) 1
end if
for i = 1 to n do
    Convert u_i^r into a vector-form delta Δ_i^r
end for
Compute the clipping threshold τ_r from {‖Δ_i^r‖_2}_{i=1}^n
for i = 1 to n do
    Δ̄_i^r ← min(1, τ_r / (‖Δ_i^r‖_2 + ε)) Δ_i^r
end for
Choose the evaluation radius q_r from a quantile of positive clipped norms
for i = 1 to n do
    Δ̃_i^r ← q_r Δ̄_i^r / (‖Δ̄_i^r‖_2 + ε)
end for
Sample T hidden tasks from P using class-balanced sampling
for t = 1 to T do
    Evaluate the baseline loss ℓ_t(w_r)
end for
for i = 1 to n do
    for t = 1 to T do
        c_{i,t} ← ℓ_t(w_r) − ℓ_t(w_r + Δ̃_i^r)
    end for
end for
Robustly standardize certificate columns to obtain C
Compute rank penalties γ from repeated coordinate subsamples of {Δ̄_i^r}_{i=1}^n
α^(0) ← α^prev
for k = 0 to K − 1 do
    z ← C^⊤ α^(k)
    π ← softmin task weights induced by z and β
    g ← Cπ − λ_r γ − 2λ_p (α^(k) − α^prev)
    α^(k+1) ← Π_{W(ρ̂)}(α^(k) + ηg)
end for
α* ← α^(K) and α^prev ← α*
Δ_agg ← Σ_{i=1}^n α_i* Δ̄_i^r
return Δ_agg, α^prev
```

Robust score calibration. The raw certificate matrix is transformed column-wise before optimization:

$$C_{i,t} = \frac{C_{i,t}^{\text{raw}} - \text{median}_j(C_{j,t}^{\text{raw}})}{1.4826 \cdot \text{MAD}_j(C_{j,t}^{\text{raw}}) + \varepsilon}.$$

This removes task-specific scale effects while preserving within-task ordering by Proposition A.5. Rank penalties are calibrated similarly. For each coordinate subsample S_s, let

$$\rho_i^{(s)} = \frac{1}{|S_s|} \sum_{j \in S_s} \left| \frac{\text{rank}_{i,j}}{n-1} - \frac{1}{2} \right|.$$

CARAT then computes a standardized rank score

$$z_i^{(s)} = \frac{\rho_i^{(s)} - \text{median}_\ell(\rho_\ell^{(s)})}{1.4826 \cdot \text{MAD}_\ell(\rho_\ell^{(s)}) + \varepsilon},$$

Table 3: Design rationale for the CARAT aggregation pipeline. Each component addresses a different ambiguity in non-i.i.d. robust aggregation.

Component	Failure mode addressed	Role in CARAT
MAD clipping and common-radius scoring	Large-norm or scale-manipulated updates can dominate validation scores, while benign non-i.i.d. clients may naturally have different update norms.	Separates the scoring copy from the final clipped update, making hidden-task evaluation less sensitive to update magnitude.
Hidden class-balanced tasks	A single global score or trusted direction can miss class-local degradation and can confuse semantic usefulness with geometric centrality.	Scores each candidate by loss reduction on multiple hidden semantic slices, producing a task-level certificate matrix.
Robust task standardization	Hidden tasks can have different baseline loss scales and difficulty levels.	Makes certificate columns comparable while preserving within-task ordering.
Coordinate-rank penalty	A client may look useful on part of the reference signal while remaining structurally extreme in update space.	Penalizes consistently outlying coordinate ranks without rewarding unusual centrality.
Capped-simplex weights	Even a high-scoring client should not receive unbounded influence in a Byzantine setting.	Limits per-client weight according to an attacker-fraction prior and bounds direct malicious mass.
Temporal regularization	Round-to-round weight oscillations can overreact to noisy probe evaluations.	Encourages stable trust allocation unless hidden-task evidence supports a change.

and aggregates subsamples through

$$\gamma_i = \max\left(0, \operatorname{median}_s z_i^{(s)}\right).$$

Thus a client is penalized only when it is consistently extreme across repeated coordinate views, not because of a single noisy subsample.

Projected optimization. CARAT initializes the projected optimizer with $\alpha^{(0)} = \alpha^{\text{prev}}$. If

$$z(\alpha) = C^\top \alpha, \qquad \pi_t(\alpha) = \frac{\exp(-\beta z_t(\alpha))}{\sum_{\ell=1}^{T} \exp(-\beta z_\ell(\alpha))},$$

then the ascent direction used in projected updates is

$$g(\alpha) = C\pi(\alpha) - \lambda_r \gamma - 2\lambda_p(\alpha - \alpha^{\text{prev}}).$$

With optimizer step size η, one iteration takes the form

$$\alpha^{(k+1)} = \Pi_{\mathcal{W}(\hat{\rho})}\left(\alpha^{(k)} + \eta g(\alpha^{(k)})\right).$$

The projection maintains feasibility throughout optimization, so every intermediate iterate satisfies nonnegativity, unit-sum, and per-client cap constraints.

Diagnostics. The implementation records learned weights, task scores, softmin task weights, rank penalties, clipping thresholds, evaluation radii, and hidden-task identifiers. These diagnostics distinguish three qualitatively different behaviors: benign but diverse clients that retain moderate weight, harmful clients that lose certificate support across hidden tasks, and structurally extreme clients whose influence is reduced mainly through the rank penalty.

A.4 Proofs and Structural Properties

The following propositions are grouped by the role they play in CARAT. Propositions A.1–A.3 characterize feasible weighted aggregation and bounded influence after clipping. Propositions A.4–A.5 characterize the hidden-task evaluation geometry. Propositions A.6–A.8 characterize the optimization objective, and Proposition A.9 records the per-round cost.

Proposition A.1 (Feasibility and support of the capped simplex)**.** *Let*

$$\mathcal{W}(\hat{\rho}) = \left\{ \alpha \in \mathbb{R}^n : \alpha_i \geq 0,\ \sum_{i=1}^{n} \alpha_i = 1,\ \alpha_i \leq \frac{1}{(1-\hat{\rho})n} \right\},$$

with $\hat{\rho} < 1$. Then $\mathcal{W}(\hat{\rho})$ is non-empty, and every $\alpha \in \mathcal{W}(\hat{\rho})$ satisfies

$$|\operatorname{supp}(\alpha)| \geq (1-\hat{\rho})n.$$

Proof. Let $u = 1/((1-\hat{\rho})n)$. Since $\hat{\rho} < 1$, we have $u \geq 1/n$, so the uniform weight vector is feasible and the set is non-empty. For the support bound, if only k coordinates of α are positive, then

$$1 = \sum_{i=1}^{n} \alpha_i \leq ku = \frac{k}{(1-\hat{\rho})n},$$

which implies $k \geq (1-\hat{\rho})n$. Therefore CARAT can suppress suspicious clients, but cannot collapse the aggregate onto an arbitrarily small subset of participants. □

Proposition A.2 (Norm bound for the final aggregate)**.** *Suppose each clipped update satisfies $\|\bar{\Delta}_i\|_2 \leq \tau_r$ and $\alpha \in \mathcal{W}(\hat{\rho})$. Then the final aggregate*

$$\Delta_{\text{agg}} = \sum_{i=1}^{n} \alpha_i \bar{\Delta}_i$$

satisfies

$$\|\Delta_{\text{agg}}\|_2 \leq \tau_r.$$

Proof. Because $\alpha_i \geq 0$ and $\sum_i \alpha_i = 1$, the aggregate is a convex combination of clipped updates. By the triangle inequality,

$$\|\Delta_{\text{agg}}\|_2 \leq \sum_{i=1}^{n} \alpha_i \|\bar{\Delta}_i\|_2 \leq \sum_{i=1}^{n} \alpha_i \tau_r = \tau_r.$$

Thus the final server step inherits the clipping radius as a hard norm upper bound. □

Proposition A.3 (Influence bound from malicious weight mass)**.** *Let $\mathcal{M}$ be the malicious clients selected in a round and let*

$$q_{\mathcal{M}} = \sum_{i \in \mathcal{M}} \alpha_i$$

be their total CARAT weight. Suppose every clipped update satisfies $\|\bar{\Delta}_i\|_2 \leq \tau_r$. Then the direct malicious contribution obeys

$$\left\| \sum_{i \in \mathcal{M}} \alpha_i \bar{\Delta}_i \right\|_2 \leq q_{\mathcal{M}} \tau_r.$$

If $q_{\mathcal{M}} < 1$ and

$$\Delta_{\mathcal{B}} = \sum_{i \notin \mathcal{M}} \frac{\alpha_i}{1 - q_{\mathcal{M}}} \bar{\Delta}_i$$

denotes the benign-only aggregate obtained by renormalizing the benign weights, then

$$\|\Delta_{\text{agg}} - \Delta_{\mathcal{B}}\|_2 \leq 2 q_{\mathcal{M}} \tau_r.$$

Proof. The first inequality follows from the triangle inequality:

$$\left\| \sum_{i \in \mathcal{M}} \alpha_i \bar{\Delta}_i \right\|_2 \leq \sum_{i \in \mathcal{M}} \alpha_i \|\bar{\Delta}_i\|_2 \leq q_{\mathcal{M}} \tau_r.$$

For the second inequality, write

$$\Delta_{\text{agg}} = (1 - q_{\mathcal{M}}) \Delta_{\mathcal{B}} + \sum_{i \in \mathcal{M}} \alpha_i \bar{\Delta}_i.$$

Therefore

$$\|\Delta_{\text{agg}} - \Delta_{\mathcal{B}}\|_2 \le q_{\mathcal{M}}\|\Delta_{\mathcal{B}}\|_2 + \left\| \sum_{i\in\mathcal{M}} \alpha_i \bar{\Delta}_i \right\|_2 .$$

Since $\Delta_{\mathcal{B}}$ is a convex combination of clipped benign updates, $\|\Delta_{\mathcal{B}}\|_2 \le \tau_r$ by Proposition A.2. Combining this with the first inequality gives the stated bound. □

Proposition A.4 (Directional invariance of certificate scoring)**.** *Suppose two nonzero clipped updates satisfy $\bar{\Delta}_i = a\bar{\Delta}_j$ for some scalar $a > 0$. Then their normalized versions in Eq.* (7) *coincide, i.e.,*

$$\tilde{\Delta}_i = \tilde{\Delta}_j .$$

Consequently, their hidden-task certificates are identical:

$$c_{i,t} = c_{j,t} \qquad \textit{for all } t \in \{1, \dots, T\}.$$

Proof. Eq. (7) rescales every nonzero clipped update to the common radius q_r. Positive collinearity is therefore removed by normalization:

$$q_r \frac{\bar{\Delta}_i}{\|\bar{\Delta}_i\|_2} = q_r \frac{a\bar{\Delta}_j}{\|a\bar{\Delta}_j\|_2} = q_r \frac{\bar{\Delta}_j}{\|\bar{\Delta}_j\|_2}.$$

Substituting the resulting equality into Eq. (8) gives identical task scores. Thus, after clipping, the certificate stage depends on direction rather than residual norm magnitude. □

Proposition A.5 (Task-wise standardization preserves within-task ordering)**.** *Fix a task t and define a standardized certificate score*

$$\hat{c}_{i,t} = \frac{c_{i,t} - m_t}{s_t + \varepsilon},$$

where $m_t \in \mathbb{R}$ and $s_t > 0$ depend only on task t and not on client i. Then for any clients i, j,

$$c_{i,t} \ge c_{j,t} \quad \Longleftrightarrow \quad \hat{c}_{i,t} \ge \hat{c}_{j,t}.$$

Moreover,

$$\hat{c}_{i,t} - \hat{c}_{j,t} = \frac{c_{i,t} - c_{j,t}}{s_t + \varepsilon}.$$

Proof. For a fixed task t, the map $x \mapsto (x - m_t)/(s_t + \varepsilon)$ is affine with strictly positive slope because $s_t + \varepsilon > 0$. Hence it preserves ordering. The difference identity follows by direct subtraction. Therefore column-wise robust standardization calibrates cross-task scale without changing the within-task ranking induced by the raw certificate scores. □

Proposition A.6 (Softmin bounds and gradient form)**.** *Let $s \in \mathbb{R}^T$ and define*

$$\operatorname*{softmin}_{\beta}(s) = -\frac{1}{\beta} \log \left(\frac{1}{T} \sum_{t=1}^{T} e^{-\beta s_t} \right).$$

Then

$$\min_t s_t \le \operatorname*{softmin}_{\beta}(s) \le \min_t s_t + \frac{\log T}{\beta}.$$

Moreover,

$$\lim_{\beta\to\infty} \operatorname*{softmin}_{\beta}(s) = \min_t s_t, \qquad \lim_{\beta\to 0} \operatorname*{softmin}_{\beta}(s) = \frac{1}{T}\sum_{t=1}^{T} s_t.$$

If $z(\alpha) = C^\top \alpha$ and

$$\pi_t(\alpha) = \frac{\exp(-\beta z_t(\alpha))}{\sum_{\ell=1}^{T} \exp(-\beta z_\ell(\alpha))},$$

then

$$\nabla_\alpha \operatorname*{softmin}_{\beta}(C^\top \alpha) = C\pi(\alpha).$$

Proof. Let $m = \min_t s_t$. Since

$$e^{-\beta m}/T \le \frac{1}{T}\sum_{t=1}^{T} e^{-\beta s_t} \le e^{-\beta m},$$

applying $-(1/\beta)\log(\cdot)$ yields

$$m \le \operatorname*{softmin}_{\beta}(s) \le m + \frac{\log T}{\beta}.$$

The limit $\beta \to \infty$ follows from the vanishing gap bound. For $\beta \to 0$, a first-order expansion of the exponential average gives

$$\frac{1}{T}\sum_{t=1}^{T} e^{-\beta s_t} = 1 - \beta\frac{1}{T}\sum_{t=1}^{T} s_t + o(\beta),$$

so the softmin converges to the arithmetic mean. Differentiating the softmin expression with respect to z_t yields the Gibbs weights $\pi_t(\alpha)$, and the chain rule gives $C\pi(\alpha)$. Hence the ascent direction is a convex combination of certificate columns, with more weight placed on tasks whose current aggregate utility is smaller. □

Proposition A.7 (Temporal prior and projected-iterate invariants)**.** *Define*

$$f(\alpha) = \operatorname*{softmin}_{\beta}(C^\top\alpha) - \lambda_r\gamma^\top\alpha - \lambda_p\|\alpha - \alpha^{\mathrm{prev}}\|_2^2.$$

If $\alpha, \alpha' \in \mathcal{W}(\hat\rho)$ satisfy $C^\top\alpha = C^\top\alpha'$ and $\gamma^\top\alpha = \gamma^\top\alpha'$, then

$$f(\alpha) - f(\alpha') = -\lambda_p\left(\|\alpha - \alpha^{\mathrm{prev}}\|_2^2 - \|\alpha' - \alpha^{\mathrm{prev}}\|_2^2\right).$$

In addition, under projected gradient ascent

$$\alpha^{(k+1)} = \Pi_{\mathcal{W}(\hat\rho)}\left(\alpha^{(k)} + \eta\nabla f(\alpha^{(k)})\right),$$

every iterate remains in $\mathcal{W}(\hat\rho)$.

Proof. The first statement follows by direct substitution: if the certificate and rank terms match, the only remaining difference in objective value is the quadratic prior term. CARAT therefore prefers, among utility-equivalent feasible weights, the solution closer to α^{prev}. The second statement is immediate from the projection operator. Hence the optimizer preserves three round invariants throughout: $\alpha_i \ge 0$, $\sum_i \alpha_i = 1$, and $\alpha_i \le 1/((1-\hat\rho)n)$. □

Proposition A.8 (Pairwise weight-transfer condition)**.** *Let*

$$f(\alpha) = \operatorname*{softmin}_{\beta}(C^\top\alpha) - \lambda_r\gamma^\top\alpha - \lambda_p\|\alpha - \alpha^{\mathrm{prev}}\|_2^2$$

be the CARAT objective, and let e_a, e_b be coordinate vectors for two clients a and b. Suppose $\alpha + \epsilon(e_b - e_a)$ remains feasible for sufficiently small $\epsilon > 0$. Define

$$\pi_t(\alpha) = \frac{\exp(-\beta(C^\top\alpha)_t)}{\sum_{\ell=1}^{T}\exp(-\beta(C^\top\alpha)_\ell)}.$$

Then moving infinitesimal weight from client a to client b increases the objective to first order whenever

$$(C_b - C_a)^\top\pi(\alpha) - \lambda_r(\gamma_b - \gamma_a) - 2\lambda_p[(\alpha_b - \alpha_b^{\mathrm{prev}}) - (\alpha_a - \alpha_a^{\mathrm{prev}})] > 0,$$

where C_i denotes row i of the standardized certificate matrix.

Proof. Let $d = e_b - e_a$. A first-order expansion gives

$$f(\alpha + \epsilon d) - f(\alpha) = \epsilon d^\top\nabla f(\alpha) + o(\epsilon).$$

Using Proposition A.6, the gradient is

$$\nabla f(\alpha) = C\pi(\alpha) - \lambda_r\gamma - 2\lambda_p(\alpha - \alpha^{\mathrm{prev}}).$$

Substituting $d = e_b - e_a$ yields exactly the displayed expression. Hence a client gains weight when its hidden-task utility advantage, measured through the current softmin focus $\pi(\alpha)$, outweighs its rank penalty and temporal-prior cost. □

Proposition A.9 (Per-round complexity)**.** *Let n be the number of clients, T the number of hidden tasks, m the number of sampled coordinates for rank scoring, S the number of rank subsamples, and K the number of projected-gradient steps. Ignoring constant factors from batch size and model width, the extra defense-side work of CARAT scales as*

$$\textit{probe-model forward passes} + O(Snm + KnT).$$

Proof. CARAT performs one baseline probe evaluation plus n candidate probe evaluations over the T hidden tasks, then computes S rank summaries over m coordinates, and finally executes K projected optimization steps over n weights and T task scores. In the prototype implementation, the probe-model forward passes dominate wall-clock time, while the rank and optimization stages are lower-order terms controlled by m, S, and K. □

Remark on the reference assumption. The object $\mathcal{D}^{\text{ref}}$ denotes the labeled server reference set, while $\mathcal{P}$ is a subsampled working pool derived from it for hidden-task construction. This distinction is part of CARAT's threat model and materially strengthens the setting relative to fully data-free robust aggregation.

B Experimental Protocols and Additional Results

This part records only empirical material that directly supports the main-text claims. It first fixes the completed CIFAR-100/ResNet18 protocol and CARAT hyperparameters, then reports final-loss results, a runtime diagnostic, and reproducibility details.

B.1 Completed Empirical Scope

The completed empirical scope is intentionally narrow. We evaluate untargeted model poisoning on CIFAR-100/ResNet18 with Dirichlet non-i.i.d. client partitions, 20% malicious clients, and four attacks: ALIE, FangAttack, MinMax, and MinSum. The comparison uses Mean, NormClipping, MultiKrum, FLTrust, FLDetector, and CARAT. All claims are based on attacked final test accuracy, attacked final test loss, and average rank. CARAT and FLTrust use matched labeled server-reference budgets, unpublished or under-review defenses are excluded from the claim set, and the result-source policy is disclosed for each table rather than hidden in aggregate statistics.

B.2 Reported CIFAR-100 Protocol

Unless stated otherwise, the CIFAR-100 comparison uses 20 clients with full participation, 200 communication rounds, 4 local epochs, learning rate 0.05, and Dirichlet non-i.i.d. partitions with $\alpha \in \{1, 0.5\}$. Legacy public-baseline rows use five random seeds where available, while regenerated FLTrust and CARAT rows use three random seeds under the matched reference-assisted protocol.

B.3 CARAT Hyperparameters

Table 5 lists the CARAT defaults used in the reported CIFAR-100 result set. These values define the reported protocol and make the server-side evaluation budget explicit.

B.4 Final-Loss Results

Table 6 reports the final test loss corresponding to Table 2. It uses the same source policy: compatible legacy public-baseline logs are reused, while matched-budget FLTrust and $T = 8$ CARAT are taken from the regenerated three-seed protocol.

Table 4: CIFAR-100/ResNet18 protocol used for the reported public-baseline result set.

Component	Setting	Role in the paper
Dataset and model	CIFAR-100 / ResNet18	Primary class-rich benchmark for the main robustness claim.
Client population	20 clients, full participation	Keeps client sampling fixed so defense differences are not confounded with participation noise.
Training schedule	200 rounds, 4 local epochs, learning rate 0.05	Shared optimization protocol for attacked runs.
Data heterogeneity	Dirichlet non-i.i.d. $\alpha \in \{1, 0.5\}$	Completed heterogeneous settings used for the main claim.
Malicious clients	20%	Attacker fraction used in the completed result set.
Attack suite	ALIE, FangAttack, MinMax, MinSum	Untargeted model-poisoning attacks used for final tables and ranks.
Compared defenses	Mean, NormClipping, MultiKrum, FLTrust, FLDetector, CARAT	Public, citable baselines used in Table 2.
Seeds and source policy	Legacy rows: five seeds where available; regenerated FLTrust/CARAT rows: three seeds	Compatible legacy logs are reused with explicit source disclosure; reference-assisted rows are regenerated with matched trusted-data budgets.
Trusted-data budget	Matched between FLTrust and CARAT	Prevents CARAT from receiving a larger labeled server-reference budget than the trusted-data baseline.

Table 5: CARAT hyperparameters used in the reported CIFAR-100/ResNet18 public-baseline runs.

Component	Default value	Purpose
Hidden-task evaluation	8 tasks, 4 probe samples per class, batch size 128	Builds repeated class-balanced certificates for candidate client updates.
Normalization and clipping	Probe-radius quantile 0.5; MAD clipping factor 2.5	Places candidates on a common evaluation radius and limits residual norm influence.
Feasible weight set	Attacker prior $\hat{\rho} = 0.2$	Caps per-client weight and prevents collapse onto a very small client subset.
Task and anomaly weighting	Certificate temperature $\beta = 12.0$; rank weight 0.20; prior weight 0.05	Balances weakest-task utility, structural rank penalty, and temporal smoothness.
Rank diagnostics	4096 sampled coordinates, 5 rank subsamples	Estimates whether a client is consistently extreme across coordinate views.
Projected optimization	60 projected steps, step size 0.5	Solves the capped-simplex weighting problem used for aggregation.

Table 6: Final test loss under four model-poisoning attacks on CIFAR-100/ResNet18 with 20% malicious clients and non-i.i.d. client partitions. Lower is better. Legacy public-baseline rows reuse five-seed logs where available; FLTrust and CARAT are regenerated under the reference-assisted protocol with three seeds and matched reference budgets.

Dirichlet α	Defense	ALIE	FangAttack	MinMax	MinSum	Avg. rank
1.0	Mean	0.0292 ± 0.0004	0.0294 ± 0.0004	0.0283 ± 0.0004	0.0283 ± 0.0006	2.2
1.0	NormClipping	0.0476 ± 0.0005	0.0501 ± 0.0004	0.0484 ± 0.0004	0.0485 ± 0.0003	6.0
1.0	MultiKrum	0.0361 ± 0.0006	0.0377 ± 0.0005	0.0295 ± 0.0005	0.0301 ± 0.0009	4.0
1.0	FLTrust (matched)	0.0363 ± 0.0016	0.0375 ± 0.0015	0.0344 ± 0.0008	0.0346 ± 0.0013	4.5
1.0	FLDetector	0.0305 ± 0.0004	0.0450 ± 0.0035	0.0283 ± 0.0006	0.0284 ± 0.0003	3.2
1.0	**CARAT** ($T = 8$)	0.0279 ± 0.0010	0.0256 ± 0.0008	0.0255 ± 0.0005	0.0252 ± 0.0009	1.0
0.5	Mean	0.0303 ± 0.0005	0.0302 ± 0.0006	0.0293 ± 0.0001	0.0293 ± 0.0004	2.2
0.5	NormClipping	0.0486 ± 0.0005	0.0517 ± 0.0011	0.0497 ± 0.0005	0.0498 ± 0.0004	6.0
0.5	MultiKrum	0.0382 ± 0.0008	0.0407 ± 0.0010	0.0306 ± 0.0006	0.0308 ± 0.0006	4.0
0.5	FLTrust (matched)	0.0396 ± 0.0005	0.0403 ± 0.0005	0.0362 ± 0.0002	0.0361 ± 0.0002	4.5
0.5	FLDetector	0.0319 ± 0.0004	0.0458 ± 0.0011	0.0294 ± 0.0004	0.0292 ± 0.0005	3.2
0.5	**CARAT** ($T = 8$)	0.0285 ± 0.0012	0.0272 ± 0.0008	0.0258 ± 0.0008	0.0258 ± 0.0004	1.0

B.5 Verification-Scale Runtime Results

Figure 3 reports a smaller CIFAR-100/i.i.d. ALIE verification run that records both round-wise accuracy and server-side runtime. In this setting, CARAT remains stable across seeds under attack but pays a clear runtime premium because each round evaluates candidate updates on repeated hidden tasks. The result is not used as the main robustness claim; it is included only to make the overhead tradeoff visible.

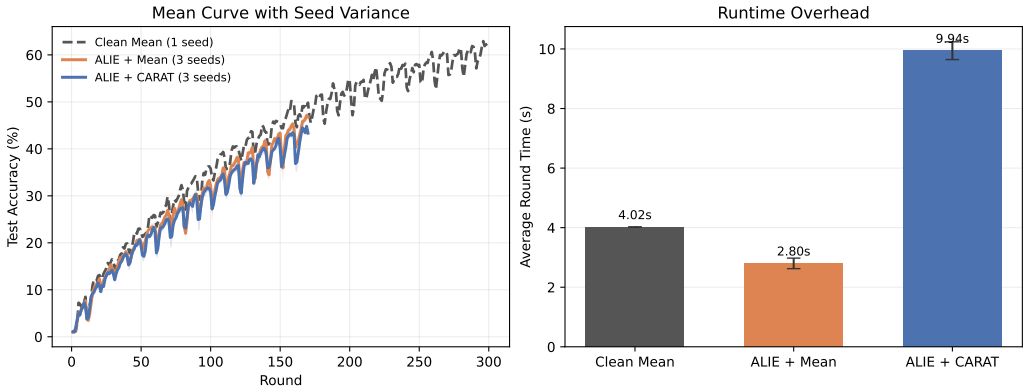

Figure 3: Verification-scale local results on CIFAR-100 under the ALIE model-poisoning attack. Left: round-wise test accuracy, where solid curves show the mean over three seeds and shaded bands denote one standard deviation; the dashed clean Mean curve is a single-seed no-attack reference. Right: average per-round wall-clock time with seed-level error bars, showing the defense overhead of repeated hidden-task evaluation.

B.6 Reproducibility, Ethics, and Availability

Reproducibility. All main reported results use the public CIFAR-100 benchmark and the fixed training protocol described in Section 5. The aggregation method is specified in Algorithm 1, with implementation defaults listed above. For anonymous review, logs, plotting scripts, and configuration files should be included in the supplementary material or an anonymized archive so that each table cell can be traced to its seed-level runs.

Compute resources. The reported public-baseline logs were generated through the repository launch scripts on a Compute Canada SLURM cluster. All runs use the same launcher, export script, and fixed configuration files. Each run writes a metadata file recording the configuration file, command line, seed, Python environment, CUDA requirement, and git commit.

Ethics. This work studies defenses against model-poisoning attacks in federated learning. The attack implementations are used to evaluate robustness and do not require access to private user data. The main practical risk is dual use: stronger attack simulations may help adversaries tune poisoning strategies. We mitigate this by framing attacks as benchmark stressors and by releasing defensive evaluation code together with the corresponding protocols.

Data availability. CIFAR-100 is a public dataset. CARAT uses a small labeled server-side reference split during evaluation; when reproducing the results, this split should be generated deterministically from the public training data under the same seed and excluded from client local training data.

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